

A GENERALIZATION STRATEGY FOR SPEECH QUALITY PREDICTION: FROM DOMAIN-SPECIFIC TO UNIFIED DATASETS

Imran E Kibria, Ada Lamba, Donald S. Williamson

Department of Computer Science and Engineering, The Ohio State University, USA
kibria.5@osu.edu, lamba.39@osu.edu, williamson.413@osu.edu

ABSTRACT

The Mean Opinion Score (MOS) has become the primary evaluation measure for speech quality assessment. However, neural networks trained on a unified collection of MOS datasets often suffer from domain diversity and their ability to generalize across diverse evaluation sets remains limited. In this work, we propose the use of Sharpness-Aware Minimization (SAM) for training on unified MOS datasets. This optimizer encourages the learning of diverse and complementary features, leading to consistent improvements across multiple evaluation sets. We validate our approach on seven training and twelve evaluation sets. Our results show that while Adam optimizer suffers from distribution shifts, SAM mitigates the effect and achieves superior generalization without a noticeable penalty in accuracy compared to models trained solely on in-domain data.

Index Terms— Mean Opinion Score, Speech Quality Assessment, Out-of-Domain, Sharpness Aware Minimization.

1. INTRODUCTION

Subjective metrics based on human feedback are considered the gold standard for quality estimation. The most commonly used metric is the Mean Opinion Score (MOS), defined as the average listener rating of an utterance on a 1–5 scale (1 = *bad*, 5 = *excellent*). Researchers have modeled this metric using deep learning, exploring various architectures [1–4], objective functions [5, 6], speech configurations for inputs, and features ranging from manually engineered representations [7] to learnable embeddings [8]. However, developing generalizable models remains challenging, as models trained on one dataset fail to generalize to others.

Generalization across MOS-labeled datasets is challenging because they are curated for different purposes. Datasets for evaluating voice conversion (VC), text-to-speech (TTS) synthesis, or speech enhancement systems typically contain synthetic speech, whereas those for analyzing transmission noise or reverberation use natural speech. MOS datasets can

also differ in language, sampling frequency, listener instructions, rating scales, and the quality of the included systems. Cooper et al. [9] have shown that listeners use the full range of the MOS scale when rating a given set of speech systems. It complicates generalization because the same systems receive lower ratings over time as advanced higher-quality systems are introduced — an effect known as the corpus effect.

Samples collected under the same experimental setup, usually examples within the same dataset, are considered in-domain. Whereas samples collected under different setups, typically examples from different datasets, are considered out-of-domain (OOD). Research in subjective speech quality assessment has been primarily limited to in-domain data. A naïve strategy for improving generalization across datasets is pooling all data prior to model training. Another approach involves pretraining a model on a single dataset and then fine-tuning it on all datasets pooled together, a method known as multi-dataset fine-tuning (MDF) [10]. However, both methods suffer from corpus effects as networks unfortunately attempt to map same quality utterances to different labels.

To address this challenge, Pieper et al. [10] proposed the AlignNet architecture, which consists of two sub-networks: AudioNet and Aligner. AudioNet is pretrained on a reference dataset to estimate MOS, while the Aligner, which is trained using MDF, maps the AudioNet MOS estimates to dataset-specific MOS scores using dataset embeddings. However, the model relies on reference dataset embeddings at test time and its performance is limited to the diversity of conditions in that dataset. Huang et al. [11] extended this architecture by modifying AudioNet to output vector representations instead of a scalar MOS value. This improvement enables finding the closest domain across available training datasets, removing dependence on a single reference dataset at test time.

Beyond architecture modifications, new loss functions for generalization have also been explored. Mittag et al. [12] proposed a bias-aware loss, which includes dataset-specific biases that map model predictions to dataset-specific MOS. These biases are modeled as learnable coefficients of a first-order polynomial and training alternates between updating model weights and bias coefficients. However, the effectiveness of this method is demonstrated primarily on synthetic datasets with limited validation across real-world, open-

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source datasets. In another study, Ragano et al. [6] have proposed contrastive regression, which results in speech representations clustered in the latent space according to their MOS labels. The training process maps each dataset domain onto a quality manifold to enhance generalization. However, this approach is computationally expensive for large batches.

Lastly, self-supervised learning (SSL) models, which were originally designed for generalization across a variety of downstream tasks, have also been explored for improving generalization across datasets. Cooper et al. [13] have shown that SSL-based models such as wav2vec2 and HuBERT achieve moderate generalization in zero-shot settings and strong performance when finetuned. Similarly, Saeki et al. [8] developed UTMOS, an ensemble of models trained on SSL representations, which improved generalization across datasets. However, SSL models consist of millions of parameters which makes them impractical for quality assessment in real-world, resource-constrained scenarios.

In this paper, we investigate the relationship between the geometry of the loss landscape and the generalization of MOS prediction networks. Prior work has shown that training deep learning models on more data improves generalization by exposing them to a broader range of conditions. However, differences in curation of MOS datasets and the corpus effect hinder networks from effectively learning across multiple datasets. To address this, we propose training MOS networks with Sharpness-Aware Minimization (SAM) [14] on a unified collection of datasets. This optimizer encourages learning of diverse input features that standard Stochastic Gradient Descent (SGD) overlooks. The resulting availability of complementary features [15], combined with improved robustness to noisy labels [14], leads to consistent gains across diverse evaluation sets. We validate our approach using seven training and twelve evaluation sets.

2. PROPOSED APPROACH

2.1. Sharpness Aware Minimization

The SAM optimizer seeks to learn a point in the weight space whose surrounding neighborhood has a uniformly low loss. In essence, it simultaneously optimizes two objectives, which includes low loss and low curvature. Foret et al. [14] have shown that with certain approximations, we can formulate the solution using alternate maximization and minimization.

In the first step, the loss is maximized within a bounded neighborhood by taking a small step in the direction of the gradients. In the second step, the training loss is minimized using gradients computed at the point of maximum neighborhood loss. Fundamentally, the two-step process minimizes the maximum neighborhood loss to converge at a flat minimum. This contrasts with conventional optimizers where only point loss is minimized using a single step optimization. The two-step SAM optimization is below.

$$\tilde{\mathbf{w}} \stackrel{\text{def}}{=} \mathbf{w} + \rho \frac{\nabla_{\mathbf{w}} L(\mathbf{w})}{\|\nabla_{\mathbf{w}} L(\mathbf{w})\|_2}, \quad (1)$$

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \frac{1}{B} \sum_{i=1}^B \nabla_{\mathbf{w}} L(\tilde{\mathbf{w}}). \quad (2)$$

Here, $\tilde{\mathbf{w}}$ and ρ denote approximation to maximum neighborhood loss and neighborhood bound (hyperparameter) respectively, whereas $\mathbf{w}$, η , B , and L denote model weights, learning rate, batch size, and training loss, respectively. In Eq. (2), the weight update follows stochastic gradient descent (SGD), but we can also use other optimizers (RMSProp, Adam, etc). Note that this two-step process becomes SGD when $\rho = 0$.

More recently, Springer et al. [15] have presented an alternate explanation to SAM's effectiveness on out-of-distribution datasets. They suggest that the optimization process implicitly balances the quality of diverse features by suppressing well-learned features and promoting underrepresented ones. Hence, complimentary features are learned that transfer better to unseen data which improves generalization.

2.2. AttentiveMOS

Although optimization is independent of network size and initialization, lightweight models trained from scratch are preferable for experimentation. SAM requires two forward and backward passes per update, and lightweight architectures help keep training feasible. Pretrained SSL embeddings can obscure the optimizer's effect on generalization, so training from scratch is important to ensure fair evaluation.

AttentiveMOS [4] is an attention-only network for speech quality assessment that aligns well with the above requirements. It takes a waveform as input and first splits it into small overlapping 2 ms frames. Each frame is then passed through a linear layer to produce embeddings, which are fed into a local block. The local block uses a Swin Transformer to extract contextual features from small groups of frame embeddings and applies a 1D max-pooling layer to reduce the number of frames. The resulting context features are then passed through a series of global blocks. Each global block is a standard Transformer that applies attention across all embeddings to capture utterance-level features. Finally, a shallow feed-forward network maps these utterance-level features to a scalar MOS prediction. In total, Attentive-MOS consists of only 86K trainable parameters which are quickly learned from scratch to predict MOS. More specific details and parameter values can be found in the original paper [4].

3. EXPERIMENTAL SETUP

3.1. Datasets

We use the SHEET toolkit [16] to download datasets for training and evaluation, comprising seven training and twelve

evaluation sets. The training sets include predefined splits from BVCC [17], SOMOS [18], SingMOS [19], NISQA [20], TMHINT-QI [21], Tencent [20], and PSTN [22]. For development, we follow the predefined split from dataset curators when available, or those provided by SHEET otherwise. Each training set is out-of-domain (OOD) relative to the others due to differences in experimental setup. The evaluation sets include test splits from BVCC, SOMOS, SingMOS, NISQA (FOR, P501, LIVETALK), TMHINT-QI, and five unseen datasets: BC-19 [23], VMC‘23 track-1a [24], track-1b [24], track-2 [25], and track-3 [26].

All datasets are publicly available and provide subjective quality labels in the 1–5 range. We train AttentiveMOS both on individual datasets and on the unified collection of all seven. Together, these datasets cover synthetic audio from TTS, VC, Singing Voice Synthesis (SVS), Singing Voice Conversion (SVC), and enhancement systems; distortions from artificial and real noise, reverberation, VoIP, transmission, and replay; six languages; and sampling rates from 8 kHz to 48 kHz. During preprocessing, each audio is resampled to 16 kHz and fixed to a duration of 20.48s.

3.2. Network Training

We use original values for all hyperparameters of Attentive-MOS as reported in [4]. Its trainable weights are optimized to reduce the standard mean squared error (MSE) between predictions and labels using the SAM optimizer, but we also optimize with Adam which was done in the original paper. The values for the learning rate, η , and neighborhood size, ρ , are set to 5×10^{-5} and 0.05 respectively. The network is trained using a small batch size of 8 for up to a maximum of 750 epochs, or less if the loss saturates. At evaluation time, we use MSE and Spearman’s rank correlation coefficient (SRCC) for analyzing model correctness and alignment to human ratings.

4. RESULTS & DISCUSSION

First, we investigate the impact of the domain diversity on the performance of Attentive-MOS with the Adam optimizer, as shown in Tab. 1. We measure the performance of Attentive-MOS, via MSE and SRCC, when trained on just one of the seven training sets (middle column) versus the unified training set (right column) that contains all seven training sets, for both the seen evaluation sets (white rows) and unseen test sets (gray rows). In the single training set experiment, only results from the best-performing training set are given.

For all seen evaluation datasets except NISQA FOR and NISQA LIVETALK, the best performance was achieved when testing Attentive-MOS with in-domain data (e.g., training Attentive-MOS with the BVCC training split resulted in the highest performance, compared to other training sets, on the BVCC test split). This matches expected behavior as the model is only being tested on data it has previously

Table 1: MSE and SRCC of Attentive-MOS with the Adam optimizer for each test set when trained on a single training set compared to the unified training set. For the single training set, only the MSE and SRCC from the best performing training set is reported. Rows highlighted in gray (BC-19, VMC‘23 track-1a, VMC‘23 track-1b, VMC‘23 track-2, VMC‘23 track-3) indicate “blind” test sets that were not used for training in either the single or unified setting. BVCC, SOMOS, SingMOS, BC-19, VMC‘23-1a, VMC‘23-1b, and VMC‘23-2 report system-level scores; FOR, LIVETALK, P501, TMHINT-QI, and VMC‘23-3 report utterance-level scores. Datasets marked with [†] are part of the larger NISQA corpus.

	Best Single		Unified	
Test Set	MSE (↓)	SRCC (↑)	MSE (↓)	SRCC (↑)
BVCC	0.336	0.717	1.047	0.642
SOMOS	0.070	0.708	0.837	0.305
SingMOS	0.153	0.566	0.273	0.068
FOR [†]	0.690	0.514	0.653	0.647
LIVETALK [†]	1.057	0.392	0.657	0.540
P501 [†]	0.933	0.627	0.980	0.609
TMHINT-QI	0.556	0.629	0.815	0.514
BC-19	0.761	0.500	1.036	0.260
VMC‘23-1a	0.446	0.377	1.219	0.078
VMC‘23-1b	0.534	0.387	1.221	0.034
VMC‘23-2	0.552	0.250	0.989	0.374
VMC‘23-3	0.610	0.595	0.860	0.382

seen. Somewhat unexpectedly, NISQA FOR and NISQA LIVETALK achieved their best performance when trained on the Tencent set. One possible reason for this behavior could be because NISQA LIVETALK contains German speech whereas NISQA Train contains only English speech. While Tencent is a Chinese-language dataset, it is the largest of the non-English datasets that contains distorted speech which may lend itself better for predicting an unseen language.

Tab. 1 demonstrates an obstacle in training networks for generalized MOS prediction. Despite the motivation to use multiple datasets for better generalization across datasets, we can see that models are not able to overcome domain diversity without intervention. Though Attentive-MOS with the Adam optimizer performs well when trained on a single set, its performance plummets (in some cases as much as 3x) indicating that the standard Adam optimization is not sufficient for gaining advantage from more data. The following experiments show that SAM better enables the model to learn from unified data compared to vanilla optimization.

Using the same training and testing setup as in Tab. 1, we now evaluate the MSE and SRCC performance when using only the Adam optimizer compared to the SAM+Adam optimizer. Figs. 1 and 2 show the MSE and SRCC values, re-

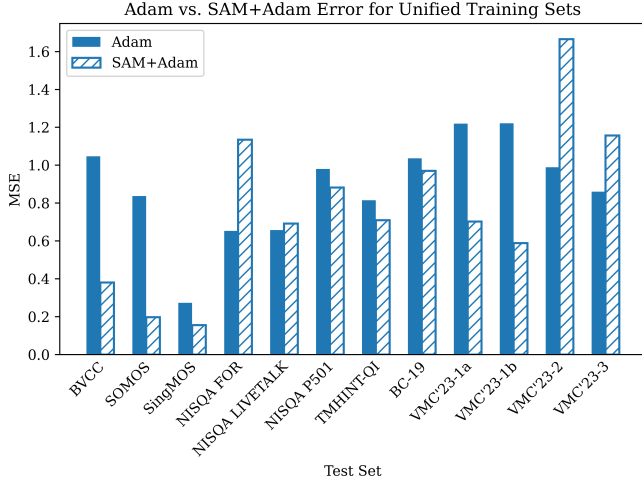


Fig. 1: MSE of Attentive-MOS trained on the unified training set with the Adam optimizer alone compared to the SAM+Adam optimizer for each test set. Shorter bars, i.e., lower MSE, indicate better performance.

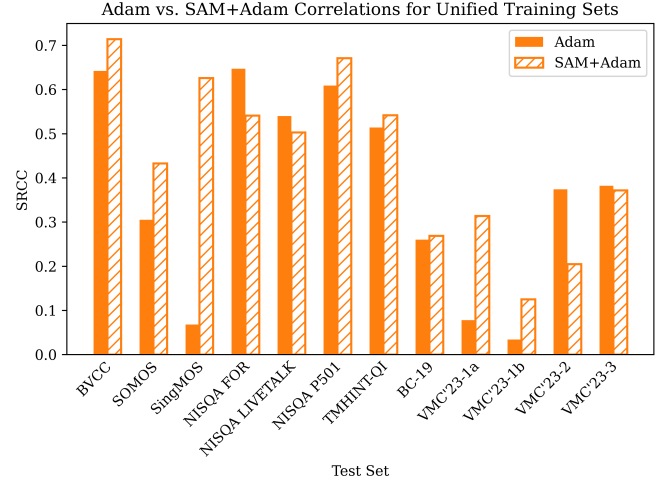


Fig. 2: SRCC of Attentive-MOS trained on the unified training set with the Adam optimizer alone compared to the SAM+Adam optimizer for each test set. Taller bars, i.e., higher SRCC, indicate better performance.

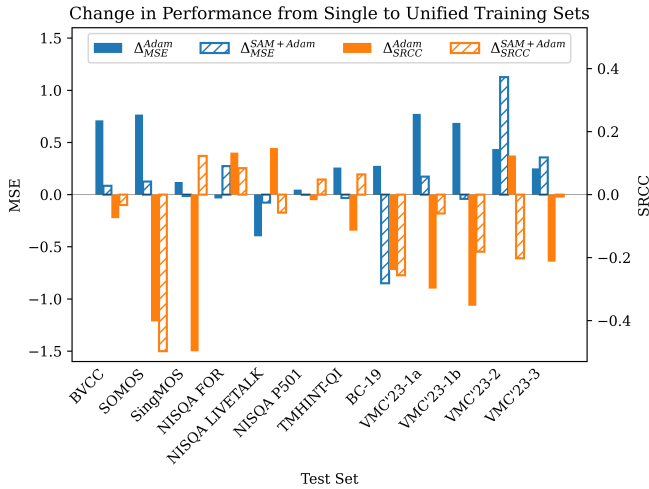


Fig. 3: Change in MSE and SRCC between the best single and unified training sets for each test set when optimizing with Adam or SAM+Adam. For MSE, lower bars indicate less penalty; for SRCC higher bars indicate less penalty.

spectively, for each test set when Attentive-MOS was trained on the unified training set. For 8 of the 12 test sets, utilizing SAM optimization lowered the overall MSE and increased the SRCC compared to Adam optimization alone. This indicates that the SAM optimizer is mitigating the corpus effect when training on OOD data, which we expand on further in Fig. 3.

Fig. 3 showcases SAM's ability to reduce the increased error when moving from a single training set to the unified training set. In Fig. 3 the bars represent the change in error, i.e., $\Delta_{MSE}^{Adam} = MSE_{Adam}^{unified} - MSE_{Adam}^{single}$. The larger the value of the Δ_{MSE} bars, the worse the MSE results were for the unified set. Conversely, the smaller the Δ_{SRCC} bars, the

worse the correlation became with the unified training set. In 8 of the test sets, the increase in error and loss of correlation from the single to the unified training set was significantly reduced with SAM+Adam optimization compared to Adam optimization alone.

In all experiments, NISQA FOR, NISQA LIVETALK, VMC'23-1a, and VMC'23-1b were the exceptions. The difference in results for NISQA LIVETALK, VMC'23-1a, and VMC'23-1b could be due to dataset language. The training sets include English, Chinese, Japanese, and Taiwanese Mandarin speech whereas NISQA LIVETALK contains only German speech while VMC'23-1a and VMC'23-1b contain only French speech. In these cases, VMC'23-1a and VMC'23-1b are not only OOD due to experimental setup and dataset curation, but due to the speech content as well. Similarly, NISQA LIVETALK is further OOD than the other NISQA test sets as it contains a language not present in the training set. NISQA FOR is the only dataset in our experiments that did not match overall trends despite being in-domain and further investigation is needed to determine the cause.

5. CONCLUSION

In this paper, we trained the Attentive-MOS network with the SAM optimizer on a unified collection of seven training sets. When evaluated on twelve test sets, the widely used Adam optimizer showed a notable degradation in performance compared to the in-domain setting, whereas SAM alleviated domain diversity, yielding lower errors and higher correlations. Beyond improving unified-dataset generalization, SAM incurred only a minimal penalty—and in some cases even improvements—relative to in-domain performance.

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